Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/08/2004
PI Name:	Thomas, John Ph.D.		
Project Title:	Quantum Coherence in Ultracold Fermionic Vapors		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jet@phy.duke.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	919-660-2508
Organization Name:	Duke University		
PI Address 1:	Physics Department		
PI Address 2:	139A Physics Bldg		
PI Web Page:			
City:	Durham	**State:**	NC
Zip Code:	27708	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	05/01/2001	**End Date:**	04/30/2005
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Maleki, Lute	**Contact Phone:**	818-354-3688
Contact Email:	lute.maleki@jpl.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNCO4GA14G		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth"s gravity		
Task Description:	The purpose of this program is a fundamental study of coherence between ground state hyperfine levels in an ultracold vapor of lithium-6 fermions contained in an ultrastable CO2 laser trap. This system permits important proof-of-principle experiments which exploit the nointeracting properties of a single component Fermi gas. The objective of this program is to broadly investigate coherence between ground hyperfine states in an ultracold fermionic gas of lithium-6 contained in an ultrastable all-optical trap. For fermions, the Pauli exclusion principle provides unique opportunities for increasing the spectroscopic resolultion. Proposed experiments include measurement of collisional shifts to determine the magnitude and sign of elastic scattering amplitudes in two-component mixtures, demonstration of double suppression of shifts and inelastic rates for coherent superposition states, demonstration of super-long coherence lifetimes for superposition states fo fermions in an ultrastable optical trap, and potentially measurement of frequency shifts arising from Cooper pairing. The ultimate coherence lifetime is expected to be limited by tthe gravitational potential and the		

	Pauli exclusion principle. The significance of the proposed program is two-fold. First, the program explores new methods for suppressing interactions by using fermionic atoms. This may impact the development of sensitive high resolution spectroscopic methods for precision measurement and atomic clocks. Further, the program utilizes and ultrastable all-optical trap with an exceptionally long lifetime and extremely low residual heating rates. This system will enable storage and study of diamagnetic atoms and molecules which could not otherwise be trapped. Second, the lithium-6 system exhibits magnetically tunable pairing interactions and is therefore an ultracold gas analog of a superconductor. Hence, it will enable precision tests of superconductivity theories in systems with widely tunable interactions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	In March of 2004, we achieved another breakthrough in observing the first evidence of superfluid hydrodynamics in a strongly interacting Fermi gas. Our experiments excite the breathing mode of the gas, by turning off the optical trap briefly. We observe these collective oscillations in a trapped, degenerate Fermi gas of 6Li atoms at a magnetic field just above a Feshbach resonance, where the two-body physics does not support a bound state. The gas exhibits a radial breathing mode at a frequency of 2837(05) Hz, in excellent agreement with the frequency of Hz predicted for a hydrodynamic Fermi gas with unitarity limited interactions. The measured damping times and frequencies are inconsistent with predictions for both the collisionless mean field regime and for collisional hydrodynamics. The observations provide the first evidence for superfluid hydrodynamics in a resonantly interacting Fermi gas.
Bibliography Type:	Description: (Last Updated:)
Articles in Peer-reviewed Journals	Thomas, J., Hemmer, S., Kinast, J., Turlapov, A., Gehm, M., and Ohara, K., "Dynamics of a highly-degenerate, strongly-interacting Fermi Gas of atoms" J. Low Temp. Phys. 134, 655 , Jan-2004
Articles in Peer-reviewed Journals	Thomas, J. and Gehm, M., "Optically Trapped Fermi Gases" American Scientist 92, pp. 238-245 , May-2004
Articles in Peer-reviewed Journals	Kinast, J., Hemmer, S., Gehm, M., Turlapov, M., and Thomas, J., "Evidence for Superfluidity in a Resonantly Interacting Fermi Gas" Phys. Rev. Lett. 92, 150402 , Apr-2004
Books/Book Chapters	Thomas, J., Hemmer, S., Kinast, J., Turlapov, A., Gehm, M., and Ohara, K. "Dynamics of a Highly-Degenerate, Strongly-Interacting Fermi Gas" in "Laser Spectroscopy, Proceedings of the XVI International Conference" Ed. P. Hannaford, A. Sidorov, H. Bachor World Scientific, 137-144 , Jan-2004
Presentation	Thomas, J., "Dynamics of a Highly-Degenerate, Strongly-Interacting Fermi Gas" Quantum Fluids Conference Aug-2003
Presentation	Thomas, J., "Exploring Superconductors and Neutron Stars on a desktop" Colloquium NC State University Oct-2003
Presentation	Thomas, J. "Exploring Superconductors and Neutron Stars on a Desktop using Ultracold Fermi Gases" Colloquium, Penn State University Oct-2003
Presentation	Turlapov, A., and Thomas, J., "Experiments with a Strongly-Interacting Fermi Gas" Southeastern Section APS Meeting Nov-2003
Presentation	Thomas, J. "Hydrodynamics in a Degenerate, Strongly-Interacting Fermi Gas" Workshop on Ultra-Cold Fermi Gases, Levico (Trento, Italy) Mar-2004
Presentation	Thomas, J., "Experiments with a Strongly-interacting Degenerate Fermi Gas" Collloquium, University of Wisonsin, Madison Mar-2004
Presentation	Thomas, J., "Experiments with a Strongly-interacting Degenerate Fermi Gas" Colloquium, Wake Forest University Apr-2004
Presentation	Thomas, J., "Hydrodynamics in a Degenerate, Strongly Attractive Fermi Gas" 2004 NASA/JPL Workshop on Physics for Planetary Exploration Apr-2004